

Xiyao Chen

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EDUCATION

Xi'an Jiaotong-Liverpool University (XJTLU) Suzhou, China | 09/2022-07/2027

- Degree: Bachelor of Science in Bioinformatics (expected in July 2027)
- Year 3 Bioinformatics Student Representative XJTLU | 2025/26
- University Academic Achievement Award XJTLU | 2026/27

RESEARCH

Bioinformatics Benchmarking of an IVT Calibration System for m6A Omics Data

Research Group Member, Director: Dr. Zhen Wei XJTLU | 06/2026-09/2026

- Benchmarked a hierarchical Bayesian m6A integration model across 2.79 million candidate sites and 14 technologies using leave-one-technology-out evaluation against statistical baselines.
- Developed Random Forest analyses to distinguish true and false m6A calls and identify technical and sequence features associated with technology-specific detection errors.
- Evaluated model robustness using orthogonal IVT evidence and writer/eraser perturbation datasets, with reproducible pipelines for cross-technology comparison.

Protein Surface Representation and Interaction Prediction Based on MaSIF

Research Group Leader, Director: Dr. Daiyun Huang XJTLU | 06/2025-09/2025

- Implemented a surface convolutional network in PyTorch to model protein-protein interactions based on an in-depth study on MaSIF.
- Conducted data splitting and model training; designed tasks about PPI binding site and affinity prediction.
- Analyzed the model's output metrics and conducted a comparative analysis with similar models.

Patterns and Drivers of Methane Emissions from Rivers and Streams Under Climate Change

Research Group Member, Director: Dr. Peng Zhao XJTLU | 06/2024-09/2024

- Collected, cleaned and analyzed GRiMeDB records spanning 5,029 sites, 24,024 methane concentration measurements, and 8,205 methane flux measurements collected from 1973 to 2021.
- Performed statistical analysis and modeling using R to identify key variables that significantly influenced methane emissions.
- Co-developed a predictive model to simulate and compare methane emission trends under various future climate scenarios.

INTERNSHIP

School of Communication and Electronic Engineering, East China Normal University

Intern in the Training and Research Group Shanghai, China | 01/2025-02/2025

- Participated in designing AI-related courses, including information collection and course outline creation.

- Processed medical data, communicated with hospital physicians regarding their needs, and assisted in advancing research projects on the intersection of AI and medicine.

Key Laboratory of Neural Regeneration, Nantong University

Research Assistant (Intern)

Nantong, China | 06/2024-09/2024

- Used R to analyze healthcare data; collected related literature
- Assisted with the translation and organization of documents related to internal statistics

ACTIVITY

XJTLU Archery Team

06/2024-Present

- Participated in routine training and competitions.
- Won the **silver medal** in 30-meter barebow mixed doubles at the Jiangsu Provincial Championships (April 2024); won the **bronze medal** in the women's 30-meter barebow event at the Shanghai Club League (March 2025).

Stock Investment Simulation Competition - Marketing Strategy Project 04/2023-05/2023

- Took charge of market research and data analysis; participated in developing marketing proposals, and assisted in formulating and implementing marketing strategies.
- Led the team to win the **first prize**.

XJTLU Summer Volunteer Activity

06/2023-08/2023

- Planned and organized information sessions at several high schools in Ningxia to introduce XJTLU.
- Managed and led the Ningxia volunteer team during the sessions.

SKILL

- Technical Skills: R, Python, Linux